

Artifact One:

Software Design and Engineering Narrative

My artifact one is from my CS210 class. I created the project on February 23rd, 2025. The original project was written in C++ and was used to track the frequency of items from an input file. The project has an object-oriented design that I carried over for my enhancement. A menu is displayed for the user giving them the option to look up an item and its frequency, along with the option to print a list of all items and the frequencies, which saves the frequency data to the file frequency.dat. The user can also print out a histogram that prints out the frequency of the items with asterisks.

I selected this item because I believe it shows my skill in software engineering and design. I also believe it shows my knowledge in both C++ and Python. Python is a skill that is important for me to have since I want to be a data engineer, and this demonstrates my skill in taking a language and converting it to Python. This artifact helped me show my ability to redesign and maintain code from one language to another. Lastly, this artifact helped to show my growth from one of my earlier courses to now in the capstone.

I believe I met the course outcomes with this enhancement, the course outcome my artifact one supported was outcome 3. This is supported because I redesigned the original code from C++ to Python while still maintaining the functionality of the original code. My enhancement improved maintainability and usability with a simpler structure, improved input validation, and input normalization when searching for items. When converting the original project from C++ to Python I decided to add methods such as `normalize_item()` and `display_menu()`, this allows to reuse the code if the project were to scale, it also improves

organization and maintainability. I also kept the object-oriented design, used cleaner code, and improved the commenting. The menu display was separated into its own method, improving organization, and readability.

One thing that I learned was that when normalizing the input, it needed to be used more than just in the `GetItemFrequency` and needed to be used throughout the code to maintain consistency even when printing the items to `frequency.dat`. I also was having trouble with the normalization when first implementing it. This was because though I got Apples and apples to work, typing in APPLES was being treated as separate values. This happened because I did not normalize the item first, I had the program `self.itemFrequency.get(item, 0)` but did not have `item = item.strip().lower()` before it. A challenge I faced was converting the object-oriented design and syntax from C++ into Python while keeping the same functionality. Object-oriented design is different between the two languages, especially when using exception handling and dictionaries. I also learned that it is more important to write meaningful code over writing comments for every line, especially on obvious lines of code.